



Operator Induced Event Reporting

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

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1.0 Introduction

Machines create a host of data that is used to monitor machine health and to point out concerns with the application. However, even though operator behaviors and human error have a significant impact on safety as well as machine and component durability, alarms related to operator behavior often do not get the required attention early on when training might still be used to correct behaviors. This then represents a lost opportunity to proactively use such information to prevent incidents and limit the adverse effect on machine durability and operating cost. This Best Practice seeks to drive a proactive reporting into the mining organization including especially operations and training departments.

2.0 Best Practice Description

Operator induced event reporting creates a means to monitor and communicate about machines being operated outside of their expected limits resulting in events being created in the on board monitoring systems. While not always black and white, e.g. a payload overload event might be driven by a faulty suspension, the ability to track these events enables mining maintenance and operations team to identify trends in operator behavior that might affect safety, machine and component health. That knowledge then creates a solid foundation to work with operators to change undesired or even harmful behavior ideally before it results in failures or incidents.

The approach mining operations take for this reporting might vary in level of detail, frequency of reporting etc. depending on preferences, availability of data or local regulations. Shorter intervals make it easier to provide almost real time feedback to operators, monitoring trends over longer periods of time can help judge the effectiveness of the actions taken to improve. However, the principle remains the same:

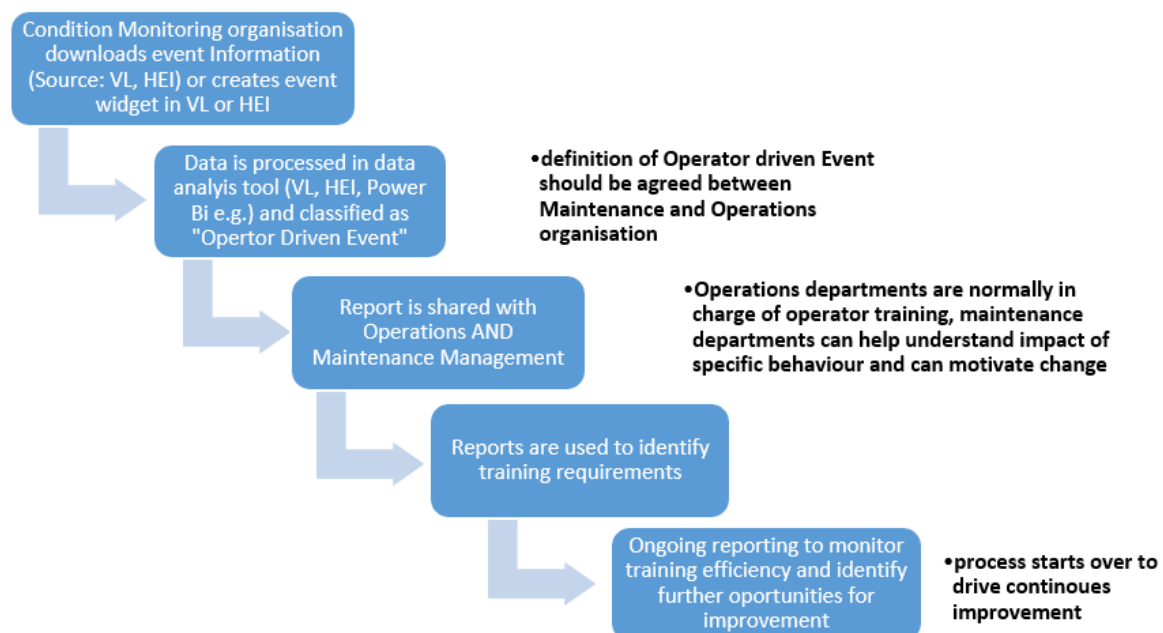


Figure 1: Suggested Process for Operator Induced Event Reporting

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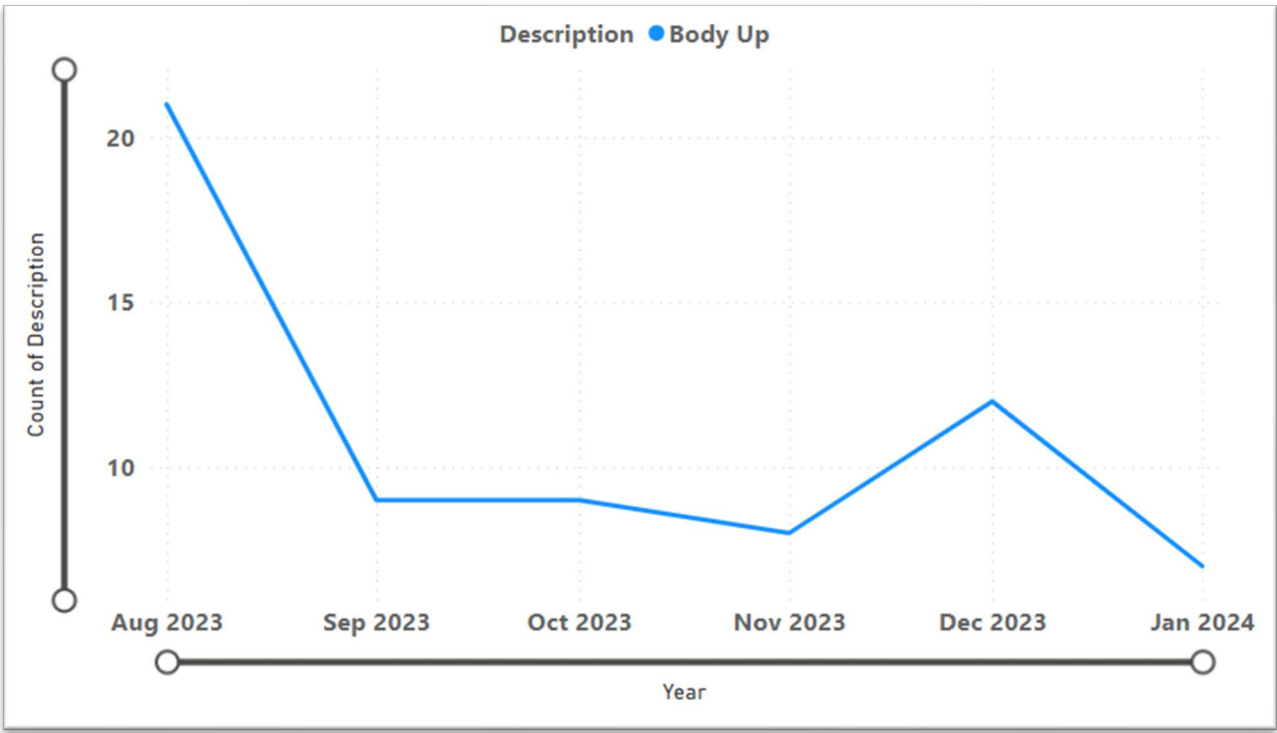


Figure 2: Tracking Events Over time allows to identify trends

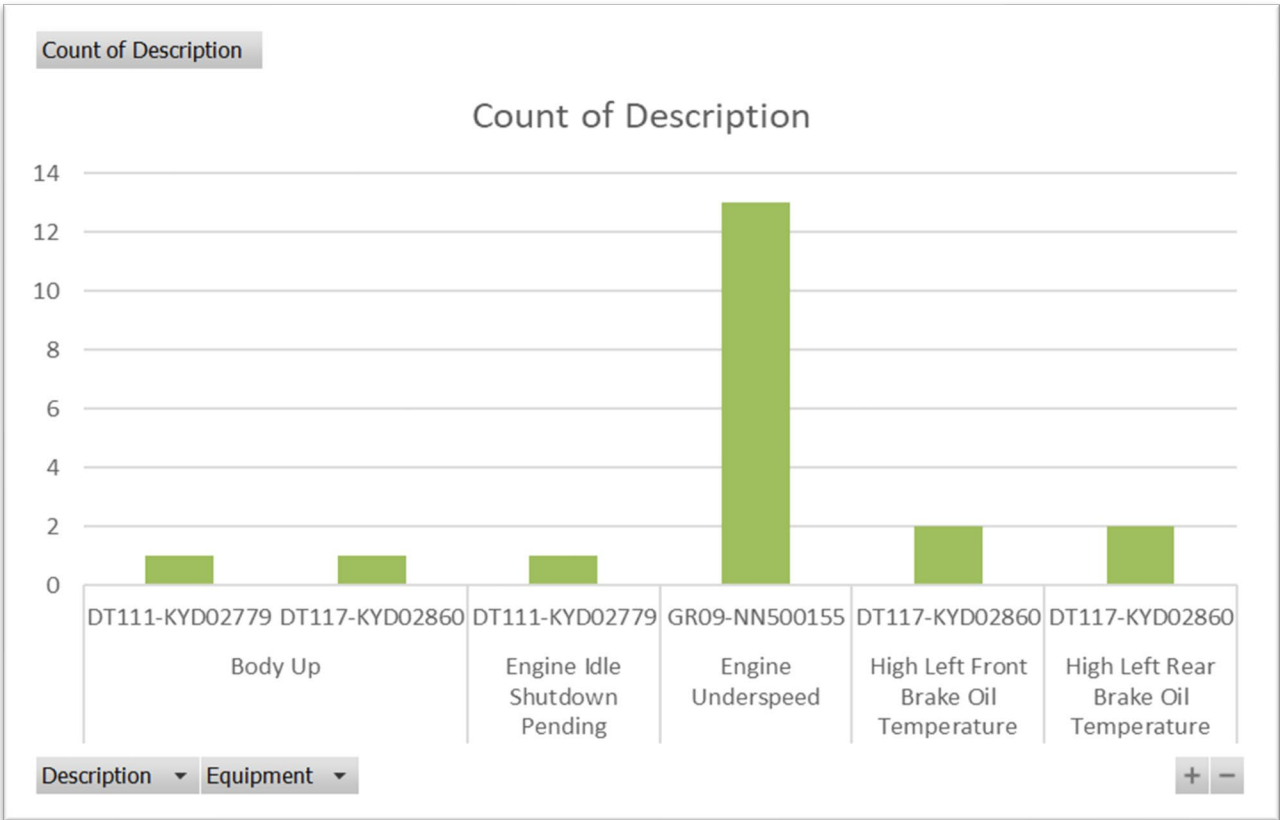


Figure 3: Example of event summary per day (Power Bi)

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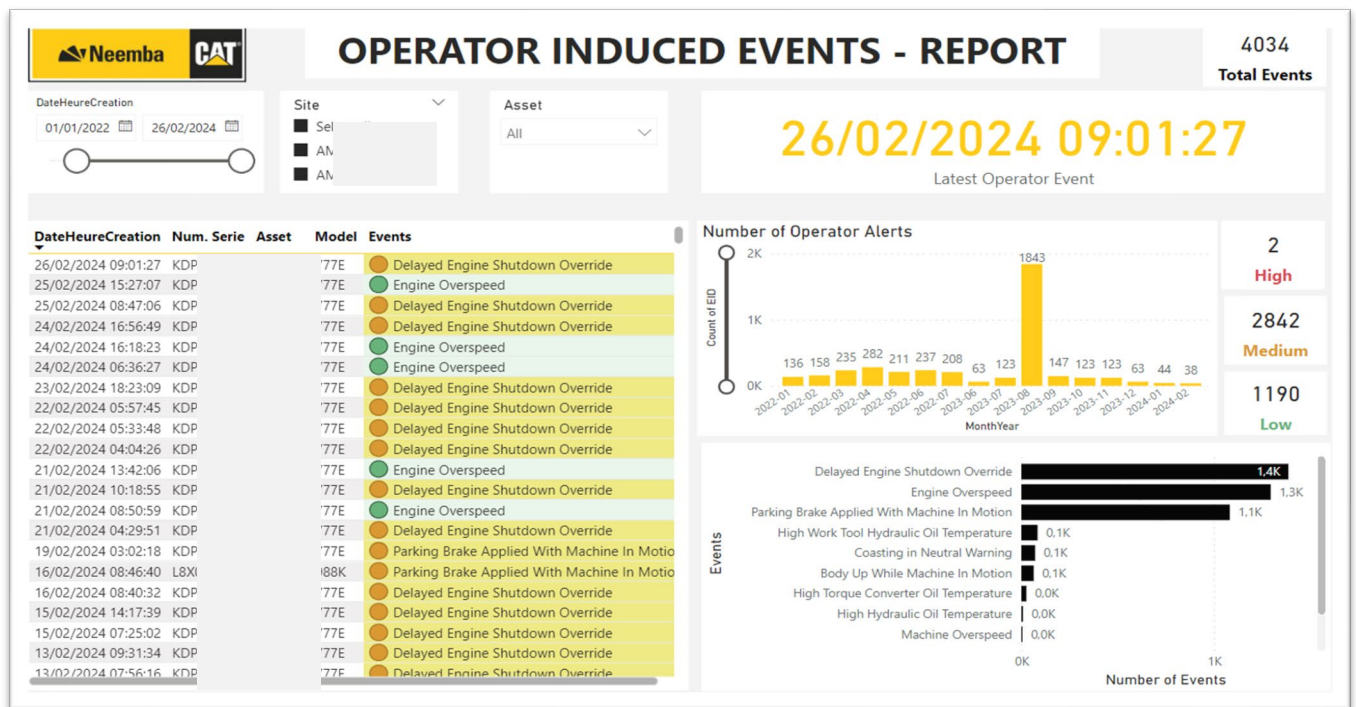


Figure 4: Example of detailed dashboard for production and maintenance department to identify trends on machine abuse.

3.0 Implementation Steps

Prerequisite for implementing this best practice is a connected fleet that reports events into the cloud on a regular basis (e.g. via MineStar Health Equipment Insights or Vision Link), the more frequent that reporting happens the more “real-time” the available data.

In addition, the following is required:

- List of events that are caused by operator abuse that can be married up with VL or HEI data using Event IDs. That list might vary depending on the models in question and based on what maintenance and operations teams on site decide to classify as “operator driven event” (example of a list below, can be amended depending on site requirements).
- Vision Link or HEI to generate dashboards directly or Power Bi or other data analysis application to link different datasets and create visualizations.
- Resource(s) dedicated to the creation of dashboards, analysis of alerts and making recommendations regarding corrective actions to improve the operations & optimize cost per tons.

Once a decision has been made to implement Operator Induced Event (OIE) reporting and the prerequisites mentioned above are in place, a decision needs to be made regarding the frequency of the report. Experience has shown that operations departments prefer daily reports to enable timely follow up with operators. This is particularly true if “operator id” is not used which makes it difficult to tie back events to individuals over longer periods.

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4.0 Benefits

OIE reporting that is shared across the mining organization makes operator driven events more visible in the entire organization. Needs for additional training can be identified on a group or individual basis and used to develop customized learning plans that drive the reduction of events. Targeted training can help improve site safety and help reduce machine operating cost.

5.0 Resources Required

- Access to machine events through the cloud (Vision Link / Health Equipment Insights).
- User proficient in creating dashboards in HEI or VL, Power Bi or other data analysis solution.
- Cross reference list to classify operator driven events (example below)
- 3W action plan to facilitate tracking of actions for training, production & maintenance departments, dealers can be added to the plan as required.

6.0 Supporting Attachments / References

Suggested list of operator Event Ids (may need to adjust based on site requirements)

Event ID	Description	Event ID	Description
e29	High Hydraulic Oil Temperature Shutdown	e362	Engine Overspeed
e30	High Hydraulic Oil Temperature Warning	e370	Screed Material Height Sensor Change During Operation
e40	Low Engine Oil Pressure Shutdown	e371	Operator Station Selection Changed while Machine in Motion
e47	Transmission Abuse Warning	e376	Gear Change with Propel Lever Not in Neutral
e48	Eng Cool Lvl	e395	Transmission Stall Condition
e49	Coasting in Neutral Warning	e482	Unexpected Left Track Forward Movement Detected
e83	Transmission Overspeed	e483	Unexpected Right Track Forward Movement Detected
e84	Machine Overspeed Warning	e484	Unexpected Left Track Reverse Movement Detected
e86	High Left Front Brake Oil Temperature	e485	Unexpected Right Track Reverse Movement Detected
e87	High Right Front Brake Oil Temperature	e490	Park Brake Applied While Shift Lever Not In Neutral
e88	High Left Rear Brake Oil Temperature	e531	Low Transmission Oil Level
e89	High Right Rear Brake Oil Temperature	e554	Low Scraper System Air Pressure
e100	Low Engine Oil Pressure Warning	e562	Steering Lever Has Not Been Aligned to Wheel Steering Angle
e106	High Front Brake Oil Temperature	e568	Unexpected Left Motor Forward Speed Detected
e107	High Rear Brake Oil Temperature	e569	Unexpected Left Motor Reverse Speed Detected
e108	Machine Upshift to Prevent Engine Overspeed	e570	Unexpected Right Motor Forward Speed Detected
e119	Low Fuel Level	e571	Unexpected Right Motor Reverse Speed Detected
e155	High Torque Converter Oil Temperature	e575	Unexpected Drive Motor Forward Speed Detected
e171	Low Engine Oil Level	e576	Unexpected Drive Motor Rearward Speed Detected
e190	Engine Overspeed Warning	e594	High Transmission Lube Oil Temperature
e191	Boom Raise Limit Reached Due to Machine in Motion	e617	Machine Lockout Active
e193	Boom Raise Limit Reached Due to Park Brake Not Engaged	e620	Starter Lockout Active
e225	Engine Overcrank	e621	Body Up
e234	Low Auto Lube Grease Level	e627	Parking Brake Applied with Machine in Motion
e237	Machine Overloaded	e642	Display Button Pressed Too Long
e242	Engine Overload	e663	Body Up while Machine in Motion
e264	Emergency Stop Activated	e667	Machine Overspeed
e265	User Defined Shutdown	e678	Ground Level Shutdown
e284	Low Brake Accumulator Pressure	e680	Access Ladder Lowered With Transmission In Gear
e289	Low System Air Pressure	e692	Rt R Brk Temp

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Suggested list of operator Event Ids ctd. (may need to adjust based on site requirements)

Event ID	Description	Event ID	Description
e358	Engine Underspeed	e698	Positive High Peak Frame Rack
e707	Lt F Brk Temp	e1388	Operator Seat Belt Unfastened While Machine is Not Idle
e709	Rt F Brk Temp	e1397	Access Ladder Lowered with Blade Raise
e712	Lt R Brk Temp	e1403	Seat Belt Unfastened with Park Brake Disengaged
e759	Brk Temp	e1560	Low Brake Pressure Due to Low Engine Speed
e777	Positive High Peak Frame Bias	e1704	High Transmission Clutch Accumulated Thermal Energy
e779	Positive High Peak Frame Pitch	e2126	Payload Overload Limit Exceeded
e784	Payload Weight Too Heavy to Zero	e2129	High Front Axle Oil Temperature
e794	Steering Systems Not Synchronized	e2130	High Rear Axle Oil Temperature
e877	High Transmission Oil Temperature	e2143	Low Engine Coolant Level
e878	High Hydraulic Oil Temperature	e2150	PTO Oil Level Low (E.C. 2810)
e888	Eng Cool Lvl	e2173	Transmission Not In Neutral at Startup
e939	Eng Cool Lvl	e2344	Park Brake Engaged While Shift Lever Not in Park
e976	Unexpected Drive Pump Signal Pressure Detected	e2916	Pyld Ovrlid Dump
e1003	High Brake Oil Temperature	e2917	Pyld Ovrlid Abuse
e1046	High Work Tool Hydraulic Oil Temperature	e3763	Pyld Ovrlid
e1152	Armrest Raised while Machine in Motion	e3803	Fuel Lvl
e1217	Delayed Engine Shutdown Override	e4701	Lo Trn Oil Lvl
e1240	Machine Operation With Rear Engine Stopped	e4790	Body Up
e1293	Transmission Internal Component Overspeed	e5747	Eng Spd
e1328	Transmission Neutral Coast Active	e8036	Fuel Tank Level Low (E.C. 2190)
e1337	Access Ladder Lowered	e13195	Coast In Ntrl
e1387	Operator Seat Belt Unfastened While Machine is Idle	e13206	Body Up

- Event ID list is attached.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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Thanks for helping to develop this best practice and put it to work at mining operations goes to Edward Nicklin, Mohamed Belal, Edwin Bwenge and Abdelhamid Hefzy of Mantrac and Maxence Hug and Papa Djibril Diagne of Neemba.

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